Felix Li

Australian Citizen | West Lafayette, IN | (765) 694-9063 | fli335491@gmail.com

Education

Purdue University West Lafayette, IN
B.S. in Aeronautical and Astronautical Engineering 2023–2027

- Cumulative GPA: 3.71/4.00
- Dean's List, Semester Honors, Presidential Scholarship

Professional Experience

Composites Manufacturing and Simulation Center Indiana, USA
Research Engineer Intern Jan 2025 – Present

Award

- Recipient of CERN CMS BTST Award for contributions to the successful repair process of the 5.3 m long BTL-Tracker Support Tube, a part of CERN's HL-LHC detector

CERN CMS Tracker Support Tube

- Key contributor to shape correction and geometric validation of a 3 m carbon-fiber composite cylinder for CERN CMS Detector structural support with a ±2 mm global geometry tolerance
- Collaborated closely with engineers to evaluate manufacturing and geometry corrective decisions, accounting for anisotropic behavior, constraint sensitivity, and deformation propagation in large composite structures
- Assisted with material and tool preparation, layup, cure planning and final assembly planning of complex composite sandwich structures, including precision epoxy bonding of multiple substructures using custom jigs and support fixtures
- Performed iterative laser scanning and surface deviation analysis for large and small structures to validate manufacturing procedures and system integration preparation

CERN CMS Tracker Support Tube Final Production Foam Cores & Prototype Tracks

- Manufactured 1m two-sided final production foam cores and prototype track sections for the CERN CMS High-Luminosity (HL) upgrade on a CNC router with optimized machining schedules and timeline
- Engineered complete CAM workflow, including material selection, custom toolpath generation, and optimization of feeds and speeds for complex 3D contoured surfaces
- Maintained strict geometric dimensions and ±0.25 mm tolerance across a 20-hour machining cycle per component
- Utilized laser scanning and VXelements to validate manufactured parts against tolerance requirements to identify improvements in the manufacturing processes

Manufacturing and Testing

- Led teams of 3–4 peers in production of more than 20 carbon fiber composite components with production timelines, spanning ply sequencing, fiber orientation, debulking, cure preparation, and oven curing
- Led end-to-end manufacturing and validation of high-precision CFRP brackets using CNC-routed tooling and laser-scanning verification, improving dimensional accuracy by 10% while achieving 0.05 mm tolerances
- Increased load bearing capacity by 20% by designing a stiffener bracket insert for CMS Detector support structure; manufactured and integrated into existing assembly
- Performed qualification, validation & mechanical characterization tests (e.g. tensile, bond strength, DCB) of composite structures for the CERN CMS Detector & Electron-Ion Collider at Brookhaven National Laboratory

Projects

ML Surrogate Modeling for Composite Structural Failure Prediction May 2026 – Present

- Developed reduced-order laminate state representations using Classical Lamination Theory (CLT), Tsai–Wu failure theory, and invariant parameters to transform discrete stacking sequences into continuous inputs
- Trained and validated XGBoost surrogate models that predict thermomechanical laminate failure directly from reduced-order laminate parameters, achieving an extrapolation R^2 of 0.985 across unseen laminates

Purdue Solar Racing Aug 2024 – Aug 2025

- Contributed to composite aeroshell surface modelling in NX and utilized PDM systems to ensure workflow efficiency
- Ansys Fluent CFD simulations for drag optimization simulating moving wheels and tire-floor contact models

Relevant Skills

- Software: MATLAB, Python (NumPy, Pandas, Scikit-Learn, XGBoost), Catia V5, Siemens NX, ANSYS Fluent, SolidWorks, Fusion360, VXelements
- Manufacturing: CAD, CAM, Composite Layup, CNC Machining, Laser Scanning, Deviation Analysis, Lean